Q(7) P, PI, PID controllers Q_2

Approx. Step function:

$$T_{sp2}(t) = \begin{cases} T_0, & 0 \le t < 5 \\ 40, & 5 \le t < 100 \\ 35, & 100 \le t < 105 \\ 40, & 105 \le t < 200 \\ 35, & 200 \le t < 400 \\ 60, & 400 \le t < 600 \\ 30 + \frac{20}{300}(t-600), & 600 \le t < 900 \end{cases}$$

$K_p = 0.352$ $\quad \tau_c = \max(0.1\tau_p, 0.8\theta_p) = 19.2\ s$

$\tau_p = 143\ s$

$\theta_p = 24\ s$

$$K_c = \frac{\tau_p + \frac{\theta_p}{2}}{\tau_c + \frac{\theta_p}{2}} = 5.0\%$$

$\tau_I = \tau_p + \frac{\theta_p}{2} = 155\ s$

$K_d = \frac{\tau_p \theta_p}{2\tau_p + \theta_p} = 11.09$